Lean Formalization of the Parameter Classification for Conditional Entropies

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This note explains how Lean 4 verifies the necessary and sufficient parameter conditions for the concrete conditional-entropy candidates studied in the complete-proof paper, Ref. [1]. The formalization covers all thirty-one numbered results: sixteen results in Sections 4 and 5 and fifteen technical results in Appendix A. For each result, the note identifies one distinct Lean theorem with the same mathematical content, allowing the reader to move directly from a statement in the paper to its formal proof. A dependency diagram also shows how the main classification is built from the underlying calculus, measure-theoretic, localization, and semiring/channel results. Every supporting result is proved using Lean and Mathlib: the development contains no unfinished proofs, project-defined axioms, or additional unproved assumptions. No previous knowledge of Lean is needed to understand the organization of the formalization described here.

1. LEAN CORRESPONDENCE

The formal development follows the complete analytic proofs in Ref. [1]. It covers the endpoint-aware Rényi parameter, real-power and entropy differentiation, signed integration, common discretization, the two-block, three-block, and Shannon localizers, exact tropical stationarity, all sufficiency and necessity branches, and the original finite semiring and conditionally mixing channel lift. The source is available at `https://github.com/robertorubboli/A-complete-characterization-of-conditional-entropies` on the repository's main branch.

Throughout this note, every numbered lemma, proposition, or remark uses the number in the complete-proof paper, Ref. [1]. That paper alone determines the organization, numbering, and conventions of the Lean correspondence.

There are two terminal conclusions. The declaration `mainClassification` says that, for every probability measure on the compactified Rényi parameter space, monotonicity of each concrete candidate is equivalent to its exact admissibility condition. Its only explicit argument is the measure being classified; it has no auxiliary hypothesis. The closed declaration `allConditionalEntropyAxioms` proves nonnegativity, zero-embedding invariance, tensor additivity, fair-bit normalization, and embedded channel monotonicity for every admissible candidate. Neither theorem assumes a project-specific axiom. In the complete-proof paper and in Lean, log is the natural logarithm and an independent fair bit has value $\log 2$.

The repository contains two complementary dependency diagrams at different levels of detail. Figure 1 is the *theorem-facing proof map*: it connects public conclusions to numbered complete-proof results and reusable proof ingredients. It is not a module-import graph. The standalone files `paper/dependency-graph.dot`, `paper/dependency-graph.svg`, and `paper/dependency-graph.png` are three formats of one detailed *Lean implementation dependency graph*; their arrows run from prerequisite to dependent.

In Figure 1, arrows deliberately point in the reverse direction, from a conclusion to ingredients used in its proof. Pale green boxes are the two public conclusions, white boxes are paper-facing intermediate results, and pale blue boxes are reusable infrastructure that is itself proved in Lean.

Table I follows the numbering of the complete-proof paper, Ref. [1]. It is a one-to-one index at the paper-facing interface: each of the sixteen numbered environments in the main text, including Remarks 5.4–5.5, has exactly one distinct canonical Lean declaration, and no declaration in the table represents two manuscript environments. All names are in the namespace `ConditionalEntropy` and are defined in `ConditionalEntropy/FullDetailsStatements.lean`. A canonical declaration may use several smaller lemmas internally. Those implementation dependencies are not

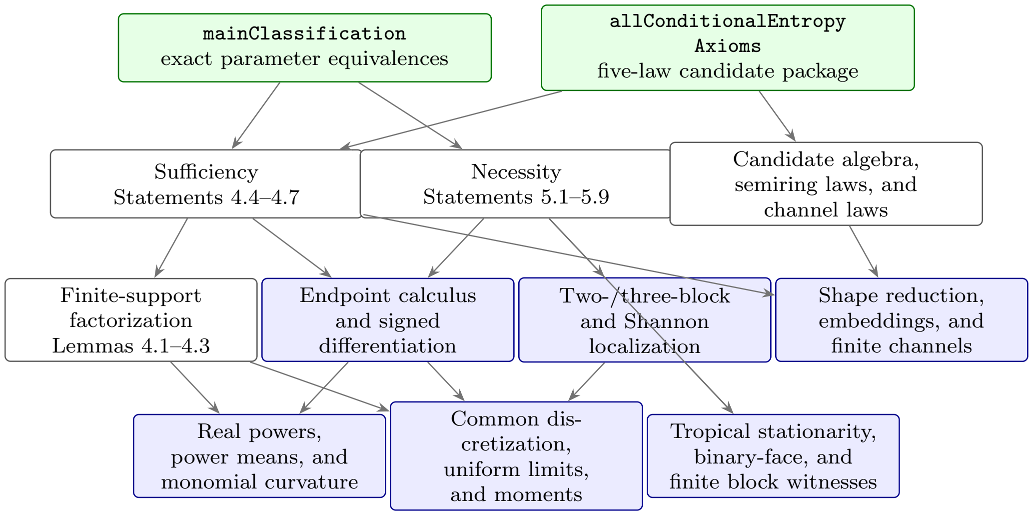

FIG. 1. Theorem-facing proof map for the parameter classification. Every displayed statement number is from the complete-proof paper, Ref. [1]. Arrows point from a result to ingredients used in its proof, the reverse of the prerequisite-to-dependent convention in the detailed Lean implementation dependency graph. Color records proof role, not confidence: green means public conclusion, blue means internally proved infrastructure, and white means a paper-facing result.

additional entries in the correspondence and are recorded separately in the source and dependency graph.

This interface-level bijection says exactly which Lean theorem a reader should inspect for each manuscript result; it does not claim that the internal proof graph is one-to-one.

TABLE I. Statement-by-statement correspondence. All numbers are from the complete-proof paper, Ref. [1].

Manuscript statement	**Lean declaration**	**Status**
Lemma 4.1: Power-mean curvature	`fullDetailsLemma4_1`	Exact
Lemma 4.2: Monomial curvature	`fullDetailsLemma4_2`	Exact
Lemma 4.3: Finite-support sufficiency	`fullDetailsLemma4_3`	Exact
Proposition 4.4: General temperate sufficiency	`fullDetailsProposition4_4`	Exact
Proposition 4.5: Negative-tropical sufficiency	`fullDetailsProposition4_5`	Exact
Proposition 4.6: Positive-tropical sufficiency	`fullDetailsProposition4_6`	Exact

Table I (continued; complete-proof numbering).

Manuscript statement	Lean declaration	Status
Proposition 4.7: Derivation sufficiency	`fullDetailsProposition4_7`	Exact
Proposition 5.1: Positive necessity	`fullDetailsProposition5_1`	Exact
Proposition 5.2: Negative Shannon atom excludes upper support	`fullDetailsProposition5_2`	Exact
Proposition 5.3: Truncated exceptional-moment bound	`fullDetailsProposition5_3`	Exact

Table I (continued; complete-proof numbering).

Manuscript statement	Lean declaration	Status
Remark 5.4: Unnormalised cone witnesses and homogeneity	`fullDetailsRemark5_4`	Exact
Remark 5.5: Compensated two-column witness	`fullDetailsRemark5_5`	Exact
Proposition 5.6: Exceptional negative branch: unique upper point, finite one-sided moments, and moment bound	`fullDetailsProposition5_6`	Exact

Table I (continued; complete-proof numbering).

Manuscript statement	Lean declaration	Status
Proposition 5.7: Negative-tropical necessity	`fullDetailsProposition5_7`	Exact
Proposition 5.8: Positive-tropical necessity	`fullDetailsProposition5_8`	Exact
Proposition 5.9: Derivation necessity	`fullDetailsProposition5_9`	Exact

Table I is the stable reader-facing interface. Each facade may call several lower-level lemmas, but those implementation dependencies do not create additional manuscript correspondences.

The repository also keeps a machine-readable 31-row audit of every numbered environment in the complete-proof document, including Appendix A, in `paper/full-details-correspondence.json`. Appendix statements are listed separately below because they provide technical ingredients rather than the main Section 4–5 conclusions. The much larger 152-row internal formalization-blueprint ledger and the editable Lean implementation dependency graph serve still different purposes and are not part of this bijection.

Table II completes the index for the fifteen numbered Appendix statements, all of which have exact facades. Row A.9 bundles joint entropy-line continuity with the exact cancellation integral for orders zero, one, and two. Thus every row in the two tables is exact.

TABLE II. Appendix statement correspondence. All numbers are from Appendix A of the complete-proof paper, Ref. [1].

Manuscript statement	**Lean declaration**	**Status**
Lemma A.1: Hessian of the power mean	`fullDetailsAppendixA_1`	Exact
Lemma A.2: Hessian and sign classification of a monomial	`fullDetailsAppendixA_2`	Exact
Lemma A.3: Continuity and uniform bound for finite-dimensional Rényi entropy	`fullDetailsAppendixA_3`	Exact
Lemma A.4: Concavity and Schur concavity in the sufficient-condition ranges	`fullDetailsAppendixA_4`	Exact

Table II (continued).

Manuscript statement	**Lean declaration**	**Status**
Lemma A.5: Measure-theoretic tools	`fullDetailsAppendixA_5`	Exact
Lemma A.6: Common discrete approximation of the Rényi parameter	`fullDetailsAppendixA_6`	Exact
Lemma A.7: Pointwise limits preserve the shape inequalities	`fullDetailsAppendixA_7`	Exact
Lemma A.8: First and second derivatives of the logarithmic power mean	`fullDetailsAppendixA_8`	Exact

Table II (continued).

Manuscript statement	Lean declaration	Status
Lemma A.9: Cancellation and continuity through the Shannon point	`fullDetailsAppendixA_9`	Exact
Lemma A.10: Continuity of the differentiated entropy at the parameter endpoints	`fullDetailsAppendixA_10`	Exact
Proposition A.11: Differentiation under the Rényi-parameter integral at fixed dimension	`fullDetailsAppendixA_11`	Exact
Lemma A.12: Dominant-block remainder	`fullDetailsAppendixA_12`	Exact

Table II (continued).

Manuscript statement	Lean declaration	Status
Lemma A.13: First two derivatives of Shannon entropy along an affine path	`fullDetailsAppendixA_13`	Exact
Lemma A.14: Second-order condition for a one-dimensional quasi-convex function	`fullDetailsAppendixA_14`	Exact
Lemma A.15: Exact finite-dimensional stationarity correction	`fullDetailsAppendixA_15`	Exact

A. Trust boundary and reproducibility

The project is pinned to Lean 4.32.2 and Mathlib 4.32.2. The release source contains no `sorry`, `admit`, project-defined `axiom`, or project-defined `opaque` declaration. The final declarations contain no unproved project hypothesis. Lean's standard logical foundations, including classical choice, propositional extensionality, and quotient soundness, remain part of the ordinary trusted base.

The single command

```
powershell -File scripts/check.ps1
```

runs the forbidden-token scan, a warning-as-error build of the integrated `CompleteCharacterization` target, the axiom audit, and a structural audit of the internal 152-row blueprint ledger. It also runs the independent full-details audit, which reconstructs all thirty-one numbers and labels from the LaTeX source, checks that canonical names are unique and in the recorded order, and asks Lean to elaborate every exact declaration. The formalization proves the classification and law package for the four concrete candidate families in the complete-proof paper's natural-log convention.

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[1] R. Rubboli, E. Haapasalo, and M. Tomamichel. *"Sufficient and Necessary Conditions for the Parameters of Conditional Entropies: Complete Proofs"*. Auxiliary document on GitHub, (2026). Available online: https://github.com/robertorubboli/A-complete-characterization-of-conditional-entropies/blob/main/paper/auxiliary-files/pdf/sections-4-5-full-details.pdf. Complete proofs for Sections 4 and 5 of the canonical manuscript.